

Faculty of Engineering

Power Electronics | First Year - Electrical Department

PROJECT NAME:

Multi-Sensor Monitoring System

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1. Project Description

The Multi-Sensor Monitoring System is an embedded systems project designed to monitor multiple environmental parameters simultaneously in real time. The system integrates several sensors with an Arduino UNO R3 microcontroller to detect hazards such as fire, flooding, and unexpected proximity threats.

A Bluetooth module enables wireless communication, transmitting live sensor data to a paired mobile device or computer for remote monitoring. A 2x16 LCD with I2C interface displays readings locally, while a buzzer and color-coded LEDs provide immediate visual and audio alerts whenever an abnormal condition is detected. This project demonstrates practical skills in embedded systems design, sensor integration, and real-time data processing — core competencies in modern electrical and electronics engineering.

2. Circuit Layout

The circuit is built on a breadboard with the Arduino UNO R3 as the central controller. Each sensor connects to designated digital or analog GPIO pins. The I2C protocol links the LCD using only two wires (SDA and SCL), while the Bluetooth module uses the hardware serial pins (TX/RX).

Output devices — LEDs and the buzzer — are wired through 220-ohm current-limiting resistors. All components share a common ground rail. Power is supplied either via the Arduino USB cable or an external battery pack. The optional water pump can be driven through a relay module triggered by the flame sensor output.

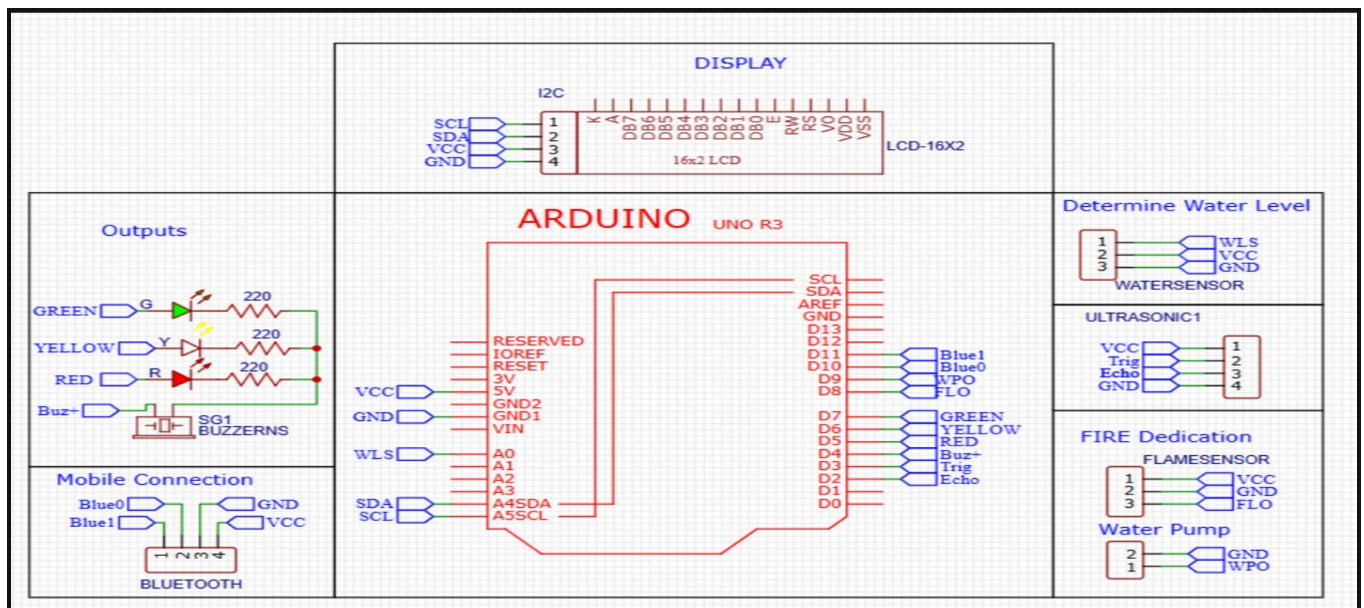


Figure 1: Multi-Sensor Monitoring System Full Wiring Schematic

3. Component List

The table below lists all hardware components used in this project:

#	Component
1	Arduino UNO R3
2	Arduino Cable for Power or External Batteries
3	Breadboard
4	Bluetooth Module (HC-05/HC-06)
5	Flame Sensor
6	Ultrasonic Sensor (HC-SR04)
7	Water Sensor
8	Water Pump (optional)
9	LEDs and (220 ohm .25w) Resistors
10	Buzzer
11	Connecting Wires
12	LCD 2x16 with I2C Module

4. Applications of the Project

The Multi-Sensor Monitoring System is versatile and can be deployed across many real-world domains:

Applications of the Proposed System

1. Integrated Smart Safety System

The combination of fire detection, distance sensing, and water level monitoring enables the development of a comprehensive safety system. It can be used in warehouses, laboratories, and industrial environments to detect fire hazards, monitor human presence, and ensure the availability of water for emergency response.

2. Automatic Fire Suppression System

The system can be designed to automatically respond to fire incidents. Upon detecting a flame, the system verifies the availability of water and activates a pump or sprinkler mechanism. This application is particularly useful in industrial facilities and storage areas where early fire suppression is critical.

3. Smart Water Tank Monitoring and Protection

Using both distance and water level sensors, the system can accurately monitor the water level in tanks and control filling or draining processes. The fire sensor adds an additional safety layer by detecting potential fire risks near the tank system.

4. Fire-Fighting Mobile Robot

The integration of distance and fire sensors allows the system to be implemented in a mobile robot capable of navigating around obstacles, detecting fire sources, and extinguishing them. The water level sensor ensures sufficient water is available before initiating the extinguishing process.

5. Intelligent Alarm System for Kitchens and Small Industries

The system can function as an early warning system by detecting fire and nearby movement. It can trigger alarms or safety mechanisms to prevent accidents in kitchens, workshops, and small industrial settings.

6. Smart Irrigation System with Fire Protection

In agricultural applications, the system can monitor water levels for irrigation purposes while also detecting fire hazards. This dual functionality enhances both resource management and safety in farming environments.

5. Software Implementation & Communication:

5.1 Mobile Application Interface:

The system integrates the Bluetooth Electronics mobile application to provide wireless communication between the Arduino circuit and the user's smartphone via the HC-05/HC-06 Bluetooth module. The application serves as an interactive interface that enables users to monitor sensor outputs in real time and receive system notifications remotely. Through this interface, sensor data including water level, distance measurements, and flame detection status are transmitted and displayed continuously, allowing effective monitoring of the system performance.

The application also supports customizable control panels, which improve user interaction by presenting sensor readings and alerts in a clear and organized manner. This enhances the usability of the proposed system and provides a practical method for remote supervision.

For further clarification, a demonstration video is attached showing the steps required to connect the application to the Bluetooth module and monitor the system data during operation using Bluetooth Electronics.

5.2 Arduino Program Implementation:

The Arduino program was developed to control the integrated monitoring system and process data from the connected sensors. The code was written using the Arduino IDE and is responsible for reading sensor inputs, controlling output devices, and handling Bluetooth communication with the mobile application.

The program initializes the connected components including the LCD display, Bluetooth module, flame sensor, ultrasonic sensor, water level sensor, buzzer, and indicator LEDs. After initialization, the system continuously checks incoming Bluetooth commands from the smartphone application to select the required operating mode.

Three operating modes were implemented in the software. The first mode activates flame detection, where the flame sensor continuously checks for fire presence and triggers an alert through the buzzer and LED indicators. The second mode enables distance measurement using the ultrasonic sensor, where the measured distance is displayed on the LCD and warning indicators are activated according to the detected range. The third mode monitors the water level by reading analog values from the water sensor and displaying the corresponding level with visual indication using LEDs.

The LCD screen provides local feedback for each selected mode, while the Bluetooth module enables wireless interaction through the mobile application. The program logic ensures that only one monitoring mode is active at a time, based on the command received from the user.

The complete Arduino source code is attached in the appendix and provided separately as a project file for further reference and implementation details

6. References:

The following resources were consulted during the development of this project:

- Monk, S. (2016). *Programming Arduino: Getting Started with Sketches* (2nd ed.). McGraw-Hill Education.
- McRoberts, M. (2013). *Beginning Arduino* (2nd ed.). Apress Media.
- Arduino Official Documentation. (2025). *Arduino UNO R3 Hardware Reference and Datasheet*.
- I2C Bus Specification. (2024). *Standard for Inter-Integrated Circuit Serial Communication*.